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ORIGINAL ARTICLE

Probiotic characteristics and survival of a multi-strain lactic acid bacteria consortium in simulated gut model

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Abstract

Dahi, a traditional yet underexplored fermented milk product from Pakistan, harbors diverse lactic acid bacteria (LAB) that have potential as probiotics. These bacteria could be used for therapeutic purposes, beneficial modulation of gut microbiota, and in the formulation of functional foods and feeds. This study aimed to isolate and characterize probiotic LAB from dahi, assess their survival in simulated gastrointestinal conditions, and evaluate their safety and probiotic potential, both phenotypically and genotypically. A total of 143 isolates from 37 samples were evaluated for probiotic traits, including acid and bile tolerance, antibacterial activity, cholesterol-lowering capacity, and antioxidant activity. The strains were also tested for antibiotic sensitivity and safety through in vitro tests and genomic analysis. A multi-strain probiotic consortium was developed and tested for enhanced functionality. Out of 143 isolates, 62 LAB strains were identified. These strains demonstrated significant survival under acidic (pH 2) and bile conditions. Antibacterial activity against pathogens ranged from 51 to 88%. The strains exhibited high cholesterol removal (up to 98%) and antioxidant activity (up to 76%). Genomic analysis revealed the presence of key probiotic-related genes, including those for acid resistance, bile salt hydrolase, and adhesion. All strains were sensitive to EFSA-recommended antibiotics and exhibited no hemolytic or DNase activity, confirming their safety. The multi-strain consortium showed superior probiotic potential and survival in simulated gastrointestinal conditions. LAB strains isolated from dahi possess strong probiotic potential, confirmed through in vitro and genomic safety assessments. The multi-strain consortium holds promise for applications.

Keywords Dahi · Probiotics · Lactic acid bacteria · Bio-preservatives · Gastrointestinal tract

Introduction

Dahi is a traditional fermented milk product that is the most used milk-based product in Pakistan (Khan et al. 2023). It is regularly incorporated into people's daily meals. Other cookery products, such as locally prepared chicken karahi, meat karahi, and dahi bhalay, among others, also incorporate it, and it serves therapeutic purposes as well. Traditionally, people prepare dahi by adding undefined mixed starter cultures or a spoonful from the previous mature batch to boiled or warmed milk (Mudgal & Prajapati 2017). Dahi is comparable to commercial yoghurt in color and appearance but is unique in taste, texture, microbiology, nutritional, and therapeutic values (Nawaz et al. 2019). It contains all the nutrients that are present in the source milk, excluding the changes triggered due to microbial growth and fermentation (Mudgal & Prajapati 2017). Microbiota of the dahi is highly diverse and reliant on a variety of factors, including the environment, utensils, and human hands. In addition to numerous lactic acid bacteria (LAB) species, dahi contain lactose-utilizing yeasts, coliforms, and spore formers (Bhattarai et al. 2016). Dahi fermentation partially breaks down complex milk proteins, including casein, making them easier to digest. This process also reduces the lactose content, which can help improve lactose intolerance symptoms in

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consumers (Mudgal & Prajapati 2017; Nawaz et al. 2016). Additionally, microbes present in the dahi produce free amino acids like methionine, lysine, and cysteine, which are beneficial for muscle and liver health (Jäger et al. 2020). Research has proven that dahi possesses antiallergic, antioxidative, anticarcinogenic, antidiabetic, antiatherogenic, lactose intolerance, immune enhancement, antimicrobial, and antidiarrheal effects (Saleem et al. 2024). Based on the properties mentioned above, it can be concluded that if the bacteria isolated from dahi are characterized, they are likely to possess probiotic potential.

Probiotics are live microorganisms that, when administered

in adequate amounts, confer a health benefit on the host (Hill et al. 2014). Due to their status as Generally Recognized as Safe (GRAS), members of the LAB are used as commercial probiotics. The global probiotic market was estimated at 77.12$ billion in 2022 and projected to experience compound annual growth rate (CAGR) of 8.2% until 2029, reflecting consumer demand for health outcomes (Share 2024). Health benefits of probiotics include but not limited to improving intestinal health, enhancing the immune response, preventing diarrhea, and improving metabolism (Das et al. 2022). LAB are primarily present in fermented foods. Hence, their isolation and probiotic characterization from artisanal fermented products have become a domain of intensive research during the last decades. The probiotic LAB obtained from artisanal fermented products can be further used in formulation of various functional foods (Azam et al. 2017). However, to qualify as an effective probiotic, the organism must be able to withstand gastric pH, endure exposure to bile salts, and adhere to the surfaces of the epithelium. Therefore, potential probiotics should be evaluated in vitro erstwhile declaring them probiotic candidates. Furthermore, it is believed that as compared to single strain, multi-strain probiotic will have high potency for beneficial impact onto the host (McFarland 2021). The present study was based on a key objective: to isolate potential probiotic LAB from still fully unexplored, artisanal fermented milk product dahi, and to formulate a multi-strain probiotic functional community. The in vitro trials and genome analysis described the isolated LAB candidates for probiotic characteristics that could be used as functional probiotics.

Materials and methods

Collection of dahi samples

Dynarex® Specimen Sterile Tamper Evident Containers (Lancaster, NY, USA) were used for sample collection. Thirty-seven samples (50 g each) of dahi were collected aseptically early in the morning, soon after the designed incubation time, from different corner dairy shops in Rawalpindi, Islamabad (Pakistan). The vendors of these shops use milk from different sources (buffalo, cow milk, or a combination of both) for preparing dahi at ambient temperature (25–40 °C). The samples were transported to the laboratory for further analysis at the same temperature. On arrival at the lab, samples were subjected to pH determination and were divided into two parts. One part was processed immediately for the isolation of LAB, and the second part was stored at − 80 °C.

Isolation of the LAB

The isolation of the LAB was grounded in the procedure used previously by Khan et al. (2023). About 1 g from each sample was taken as an initial dilution and was serially diluted using 0.85% (w/v) NaCl. From each dilution, 100 µL was plated on MRS, M17, skim milk, and tryptic soy media. Media were purchased from Sigma-Aldrich (Louis, MO, USA). The plates were incubated aerobically and anaerobically at 37 °C for 24–72 h. Morphologically different colonies were purified into single isolated colonies via sub-culturing onto respective media plates. The pure cultures were stored at 4 °C for further analysis, and the stock cultures were preserved at − 80 °C in 30% sterile glycerol. Unless otherwise specified, the same medium on which the LAB were isolated were used for the rest of the analysis.

Identification of LAB

Initial identification of 143 isolates was based on Gram's staining, catalase-oxidase activity, and motility tests. The isolates were then assessed for acid production, lactose utilization, and curd formation as described previously (Khan et al. 2023). The isolates were evaluated for mode of fermentation (homo/hetero fermentation) and the ability to ferment different types of sugars.

Molecular identification of the 88 potential LAB was confirmed through the ON-Rep-Seq method (Krych et al. 2019). DNA was extracted from freshly grown colonies through the InstaGene™ Matrix (Bio-Rad Laboratories, California, USA), visualized through 2% gel, and quantified through the Qubit™ 4 fluorometer (Thermo Fisher Scientific, Massachusetts, USA). The samples were prepared for ON-Rep PCR cycles. After PCR, 10 µL of each sample was used to construct the sequencing library, which was sequenced through MinION (Oxford Nanopore Technologies, Oxford, UK). The sequenced (ON-Rep-Seq) data was demultiplexed in fastq for the generation of read length count profiles (LCps). Reads within each peak were grouped with VSEARCH, modified with Canu, and sorted taxonomically.

Evaluation of probiotic properties of LAB

Survival in acids and bile salts

Survival of the LAB in low pH and bile salts was evaluated using methods described previously (Albano et al. 2018; Choi et al. 2018). Fresh culture of the LAB was inoculated in media broth set to pH values 2 with hydrochloric acid (HCl). The media were supplemented with 0.45% Oxgall (Sigma-Aldrich, USA) for evaluating survival in bile salts. The isolates were incubated at 37 °C for 0, 2, and 4 h and cultured for enumeration using the respective media. After incubation, the percent survival of the isolates was determined using the equation:

$$\text{Survival (\%)} = \frac{\text{CFUT1}}{\text{CFUT0}} \times 100 \tag{1}$$

where CFU is presenting colony-forming units, T1 presents CFU at the time, and T0 presents CFU at time 0.

Survival in simulated gastrointestinal fluids

Individual strains of the LAB were initially assessed for survival in simulated intestinal fluids. Simulated fluids (SF), i.e., salivary fluids, small intestinal fluids, and gastric fluids, were prepared according to the method published previously (Khan et al. 2023). Then, 1.3 mL of each SF was taken, to which 0.2 mL of LAB suspension twice washed with phosphate-buffered saline (PBS) was added. The samples were then incubated at 37 °C for 4 h. 100 μL of the samples was taken at 0-, 2-, and 4-h intervals and were plated onto the agar plates and incubated at 37 °C for 24 h. The percent survival was calculated using the formula (1).

Evaluation of antibacterial properties

The antibacterial effect of LAB was assessed through measurement of optical density (OD) against *Bacillus subtilis* subsp. subtilis ATCC 19659, *Escherichia coli* ATCC 25922, *Salmonella enterica* subsp. enterica ATCC 27870, *Pseudomonas aeruginosa* ATCC 27853, *Staphylococcus aureus* subsp. aureus ATCC 6538, and *Streptococcus pneumoniae* ATCC 49619. Optical density provide a precise, continuous scale for detecting antimicrobial effects, eliminating diffusion issues and offering greater sensitivity than zone inhibition assays, especially at lower concentrations (Fredborg et al. 2013). The fresh overnight culture of LAB was initially centrifuged (5000 × g, 10 min, 4 °C). The supernatant was filtered with a 0.2-µm syringe filter and divided into two equal parts. One part was used as it was extracted through centrifugation, i.e., the cell-free supernatant (CFS), and its pH was 3.7 ± 0.14. For the second part, the pH was adjusted to 6.5 using NaOH/HCl to remove the assumed outcome of organic acids present in the filtrate and to assess antimicrobial compound production. This adjusted supernatant is referred as n-CFS. Both parts were then stored at − 20 °C for further analysis.

Then, 50 μL of pre-prepared filtrate, 50 μL of the indicator strain, and 200 μL of Luria Broth (LB) were poured into a 96-well plate. The LB inoculated with pathogenic suspension was used as a positive control, and broth deprived of any inoculation was used as a negative control. The well-plate was incubated at 37 °C for 24 h, and OD was taken at 600 nm. The total percent pathogen inhibition was determined with the following equation (Georgieva et al. 2015).

$$\text{Inhibition (\%)} = \frac{\text{OD control} - \text{OD test sample}}{\text{OD control}} \times 100 \tag{2}$$

Antibiotic sensitivity

Antibiotic susceptibility of the isolated LAB was determined through minimum inhibitory concentration (MIC) assay against various antibiotics recommended by European Food Safety Authority (EFSA) following standard protocol (Ren et al. 2018). The susceptibility was assessed using ampicillin, vancomycin, gentamicin, kanamycin, streptomycin, erythromycin, clindamycin, tetracycline, and chloramphenicol purchased from Sigma and recommended by EFSA (Rychen et al. 2018). The MIC cut-off values noted and compared with the EFSA guidelines. The results are expressed as susceptible, intermediate, or resistant.

Cell surface hydrophobicity and auto-aggregation

Cell surface hydrophobicity (CSH) and auto-aggregation of LAB was assessed in vitro through the assays explained previously (Dlamini et al. 2019). For CSH, fresh cultures were centrifuged for 10 min at 8,000 × g and 4 °C, cleansed two times with saline (0.85% w/v), and re-suspended in saline, subsequently measured its absorbance (A0) at 600 nm. A 3 mL volume of bacterial cells was combined with 1 mL xylene and kept static at 37 °C for 1 h. That resulted in the appearance of two separate phases, i.e., organic and aqueous. The aqueous phase cautiously took off and recorded its absorbance (AT) at 600 nm. Percentage CSH was calculated via reduction in absorbance through the following formula:

$$\text{CSH (\%)} = 1 - \frac{\text{AT}}{\text{A0}} \times 100 \tag{3}$$

The AT presents absorbance at a time, and A0 shows the absorbance at time 0.

For auto-aggregation, after centrifugation and cleansing the sample was left to stand for an hour at 37 °C, and the upper phase taken and measured for absorbance at 600 nm. The percent auto-aggregation was calculated using the formula (3).

Exopolysaccharide production

To evaluate exopolysaccharide (EPS) production ability of the isolates, a modified Skim Milk agar with 10% lactose was used as described previously (Pailin et al. 2001). Fresh culture of the LAB was streaked on the plates and incubated at 37 °C for 24 h. After incubation, the plates were observed for EPS production. Colonies that appeared mucoid were considered as EPS-producing isolates.

Bile salt hydrolase activity

A method described Tanaka et al. (1999) was used to detect the bile salt hydrolase (BSH) activity of LAB. Agar media were supplemented with 0.5% (w/v) bile salts of taurodeoxycholic acid and glyco-deoxycholic acid (Sigma-Aldrich) along with 0.37 g/L of $CaCl_2$. Overnight culture of the LAB was spotted on the plates and incubated at 37 °C for 24–72 h. Precipitation below and around the spots indicated positive BSH activity of the isolates.

In vitrocholesterol lowering ability

To assess LAB isolates' in vitro cholesterol-lowering ability, the method described by Madeeha et al. (2016) was followed. LAB were grown in respective broth media for 15 h, and 100 µL of the bacterial suspension was then transferred to 9.9 mL of fresh broth. The 0.4 mg/mL (0.1 mL) cholesterol was added into the test tube to make available 0.04 mg/mL to the isolates, and they were incubated at 37 °C for 24 h. After incubation, a 0.1-mL sample was taken from each test tube and added to the test tube with 9.9 mL of $FeCl_3$-acetic acid solution. The tubes were vortexed for 15 min and subsequently centrifuged at 8000 × g for 10 min. Then, 5 mL of the supernatant was taken in a separate sterile test tube, and 3 mL of H_2SO_4 was added to it and vortexed for 30 s. The OD of the sample was measured at 560 nm after 30 min of incubation in the dark.

The reduction was measured with the following formula:

$$\text{Cholesterol removal (\%)} = \frac{\text{cholesterol removed}}{\text{Total cholesterol in media}} \times 100 \tag{4}$$

Decarboxylase activity

For evaluating the decarboxylase activity of LAB, the method used by Bover-Cid and Holzapfel (1999) was followed. Tyrosine, histamine, putrescine, and arginine each 0.5% individually, along with bromocresol purple as an indicator were added to culture media. Then, 100 µL of fresh culture of the LAB was transferred to the test medium and incubated at 37 °C for 72 h. After incubation, color change was observed as positive, while no color change was considered negative.

Antioxidant ability by radical scavinging ability

Antioxidantint activity of the LAB was performed following the method described by Yan and Polk (2020). To determine the activity, 2 mL of 2,2-diphenyl-1-picrylhydrazyl (DPPH) solution (0.4 mM) in methanol and 2 mL of LAB suspension were mixed in a sterile microtube and incubated at 37 °C for 30 min in the dark. The ascorbic acid (1 mg/mL) was utilized as a positive control. After 30 min of incubation, the tubes were centrifuged for 10 min at 8000 × g and 4 °C. The antioxidant activity was determined by recording the absorbencies of the supernatant at 517 nm and then using the following formula:where AS and AC present sample and distilled water respectively.

$$\text{Antioxidant activity (\%)} = 1 - \frac{\text{AS}}{\text{AC}} \times 100 \tag{5}$$

Safety assessment of LAB strains

Hemolytic activity

The hemolytic pattern of LAB was observed through the method described by Angmo et al. (2016). The isolates were streaked onto blood agar and incubated at 37 °C for 48 h and were observed for α, β, and γ hemolysis.

DNase activity

To evaluate the DNase activity, the method reported by de Almeida Júnior et al. (2015) was followed. LAB were streaked onto DNase Media (Sigma-Aldrich) and incubated at 37 °C for a duration of 48 h. Following incubation, the plates were immersed in a solution of 0.1% toluidine blue and allowed to sit for of 5 min. The isolates that tested positive for DNase exhibited a distinct area devoid of growth surrounding the streak line of the colonies, but the isolates that tested negative for DNase did not show any such area.

Technological characterization of LAB isolates

The enzymatic potential of LAB strains

For evaluation of proteolytic activity of LAB milk agar plates were prepared according the method described by de Giori and Hébert (2001). The plates were streaked with LAB and incubated at 37 °C for 48 h. After incubation, clear zone around the streak lines was observed for positive results.

For evaluation of the lipolytic activity, Tween 80 was used as a lipid substrate in the media as described by Leuschner et al. (1997). Methyl red was added as an indicator in the media before pouring into the plates. LAB were streaked on the media plates and incubated at 37°C for 24 h. Change from red to yellow color was observed for lipolytic activity.

The method described by Lee et al. (2001) was used for amylolytic activity. Surface dried plates of starch agar (Sigma-Aldrich) were streaked with LAB culture and incubated at 37 C for 48 h. These plates were then flooded with gram's iodine for 15 to 30 min. A clear zone around the streak lines was observed for cellulolytic isolates.

Carboxymethyl cellulose (CMC) media was prepared according to the protocol described by Hankin and Anagnostakis (1977). LAB was point inoculated on the plates and was incubated at 37 C up to 5 d. The plates were stained with 1% Congo red and rinsed with NaCl. The zones that appeared around the colonies were considered positive results.

Survival of LAB isolates at a different salt concentration

The survival of LAB at different salt concentrations such as 4 and 6% for *Lactobacillus* species and 2 and 4% *Lactococcus* species, respectively, was evaluated for 2 and 4 h at 37 °C following the method described by Somashekaraiah et al. (2019). The viability of the isolates was assessed through the plate count method (CFU/mL).

Survival of LAB isolates at different temperatures

The growth of the isolates was checked at a low temperature of 15 °C, and a high temperature of 45 °C. The growth of the isolates was evaluated by plate count assay (CFU/mL), and the results are displayed as $\log_{10}$ (Somashekaraiah et al. 2019).

Genome sequencing and assembly

DNA was extracted from fresh pure colonies using the InstaGene™ Matrix (Bio-Rad Laboratories, Inc, NY, USA) and Qubit 2.0 fluorometer (Invitrogen, Carlsbad, CA, USA) was used to check the quality of the extracted DNA. Libraries for high-throughput sequencing were made by the commercially available Vazyme TruePrep™ DNA Library Prep Kit V2 for Illumina (Vazyme Biotech, Nanjing, China). Sequencing was performed on an Illumina HiSeq-2000 sequencing platform (BGI-Shenzhen, Shenzhen, China). The trimming of the sequencing reads was performed using the Trimmomatic version 0.38 (Bolger et al. 2014). High quality reads were filtered using BWA and then aligned to the reference sequences. The assembly of the high-quality sequencing reads was performed using Velvet v. 1.2.10 assembler. The de novo assembly results were evaluated using the Quast software (Gurevich et al. 2013).

Genome analysis

The genomes were annotated through the Prokaryotic Genome Annotation Pipeline (PGAP) and Rapid Annotation System Technology (RAST) (Aziz et al. 2008). Functional annotation was performed by using EggNOG-mapper tool e-mapper v.2.1.6–25-g1502c0F (Huerta-Cepas et al. 2019). Bacteriocin genes were detected in the genomes through the online tool BAGEL4 http://bagel4.molgenrug.nl with default settings. Antibiotic resistance genes acquired within genomes were identified using comprehensive antibiotic resistance database (CARD v3.2.9) (Alcock et al. 2020). Putative virulence factors and pathogenicity islands were detected using ResFinder v4.5.0 and Islandviewer4 https://www.pathogenomics.sfu.ca/islandviewer (Bertelli et al. 2017). Pathogenicity of the genomes were determined using PathogenFinder v1.1. Putative prophage insert regions were identified using the PHage Search Tool Enhanced Release (PHASTER) (Arndt et al. 2016). Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) and CRISPR-associated genes (Cas) were detected using CRISPRCasFinder (Couvin et al. 2018). Secondary metabolite biosynthesis gene clusters were identified using antiSMASH bacterial v. 7.1.0 (Blin et al. 2023). Carbohydrate-active enzymes (CAZymes) within the genomes were identified using the CAZy database. An online database (iLABdb, https://www.imhpc.com/iLABdb/#/Map?type=%5Bobject%20Object%5D) was used to investigate the polysaccharide metabolism and short-chain fatty acid (SCFA) biosynthesis potential of the LAB.

Development of multi-strain consortium

A multi-strain probiotic consortium (MSPC) was developed from the strains. Two media were selected: skim milk (SM) and tryptic soy broth (TSB). SM was chosen because the strains were of dairy product origin, hypothesizing optimal growth and interaction in a similar environment. TSB, a general purpose medium, was used to assess growth under standard conditions. Growth was significantly better in TSB, which was selected for further experiments (Khan et al. 2024). Briefly, in the first step of consortium development, strains that may have a possible antagonist effect, such as *Enterococcus faecium, E. lactis, Lactococcus lactis,* and *Lactobacillus acidophilus,* were co-cultured with all the remaining strains individually to know that they are compatible with each other and do not inhibit each other through cross-streak method (Balouiri et al. 2016). Following addition method, all the strains were gradually combined in the second step after their 16 h of incubation. The vitality of the strains was assessed through 16S rDNA amplicon sequencing of the respective CFUs after 16 h of incubation (Imran et al. 2010). To ensure uniformity in the composition, 100 µL suspension from individual strains with 10^7 CFU/mL was mixed. The final consortium was again diluted to 10^7 CFU/mL to maintain a uniform composition using the McFarland standard. The MSPC was grown in Tryptic Soy broth, with the pH adjusted to 6.5, and incubated under anaerobic conditions using an anaerobic jar (BBL, Gas Pak Anaerobic Systems) for 24 h. MSPC was evaluated for its survival using a simulated model of the small intestine (TSI) through 16S rDNA amplicon sequencing from colonies. An overview of the model, description and sampling points are presented in the in the supplementary material (Supplementary file 1). The MSPC was assessed for CSH, auto-aggregation, antioxidant activity, antipathogenic activity, and in vitro cholesterol lowering ability using same methods that are explained earlier in the manuscript for individual strains.

Statistical analysis

All the experiments were conducted randomly in triplicate, and the statistical analysis was performed using R version 4.4.1. Significant differences ($p < 0.05$) between the strains were determined by one-way analysis of variance and Duncan's multiple range test.

Results

Identification of LAB isolates

Initially, 143 distinct bacterial isolates were selected from 37 samples. Out of these, 88 isolates were Gram-positive, non-motile, and catalase-oxidase negative. These isolates were also positive for acid production, milk curding, and lactose utilization. Using the ON-Rep-Seq, 62 isolates were identified as LAB (Table S1). The isolates were subjected to characterization for the desired probiotic traits, e.g., resistance to low pH, bile salts, survival in intestinal fluids, antipathogenic potential, sensitivity to EFSA recommended antibiotics, cell surface hydrophobicity, auto-aggregation, BSH activity, and cholesterol lowering potential, 19 isolates proved to be potential probiotics (Fig. 1). Data of these isolates as single strains and in multi-strains consortium is presented here in the manuscript. Mode of fermentation and the ability of the isolates to ferment different sugars are presented in Table S2**.**

A Phylogenetic tree illustrating the relationships among strains of LAB. Red color shows strains isolated in this study, while other branches represent similar strains. **B** Heatmap of average nucleotide identity (ANI) and genome-to-genome distance calculator (GGDC) percentages for the studied strains compared to reference genomes. The metrics provide insights into the genetic similarity and distances among the strains.

In vitro determination of potential probiotic properties

Acid and bile tolerance

The ability of bacterial strains to survive in an acidic environment is most likely to depend on the acidic pH of the stomach. The isolates were found to have a survival rate above 50% in the acidic environment (pH 2) after 4 h of exposure. However, among the tested isolates *Limosilactobacillus fermentum* QAULFN51, *L. fermentum* QAULFN54, and *L. fermentum* QAULFN64 had statistically significant survival ($p < 0.05$) in the low pH after 4 h of (Fig. 2A). Similarly, survival in bile salts helps in the characterization of the isolates for their survival in the small intestine. The isolates had more than 50% survival in bile salts. However, *E. lactis* QAUEFNA13, *E. faecium* QAUEFNN4, *L. fermentum* QAULFN54, and *L. fermentum* QAULFN55 had significantly ($p < 0.05$) high survival in 0.45% of bile salts (Fig. 2B).

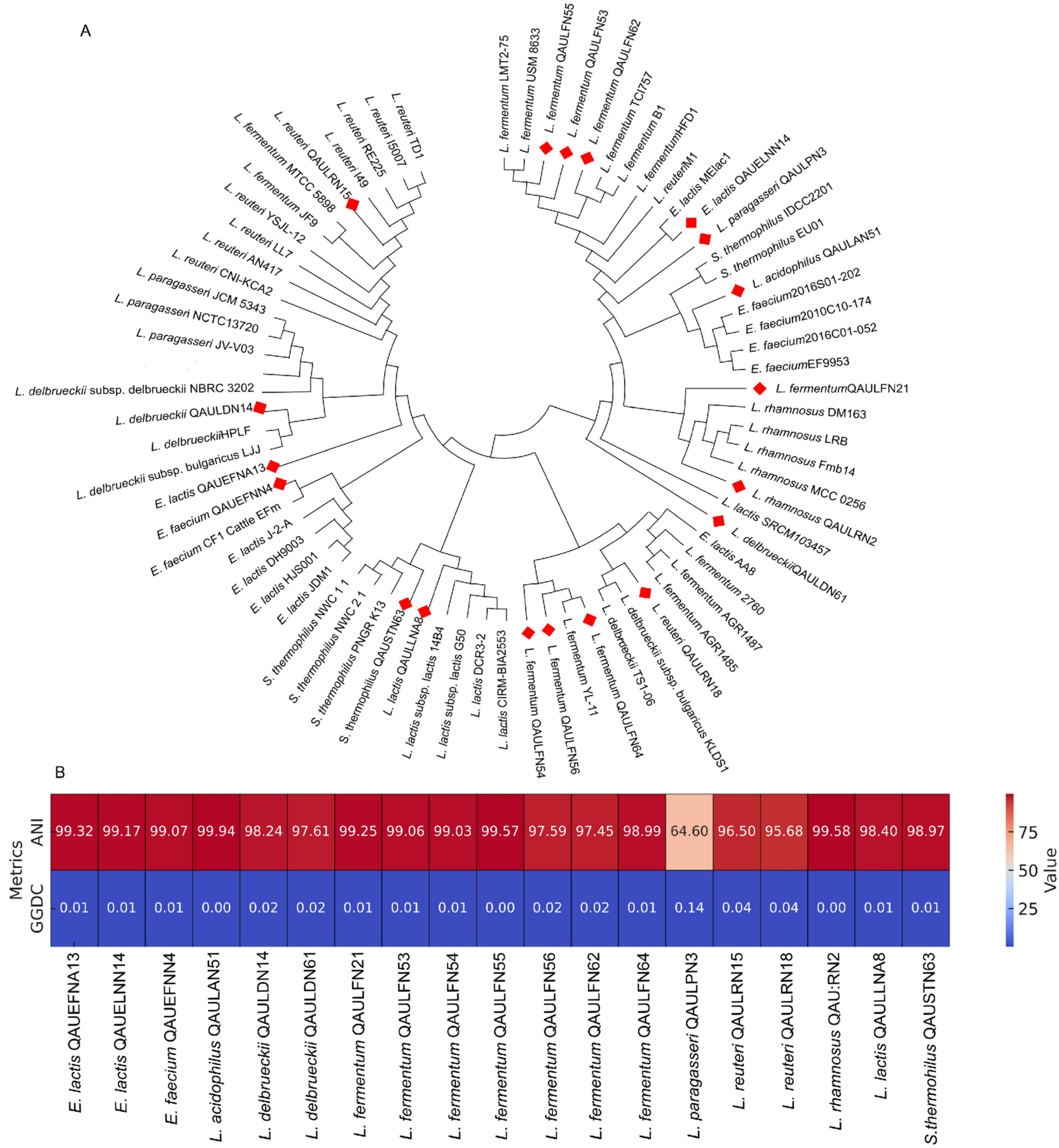
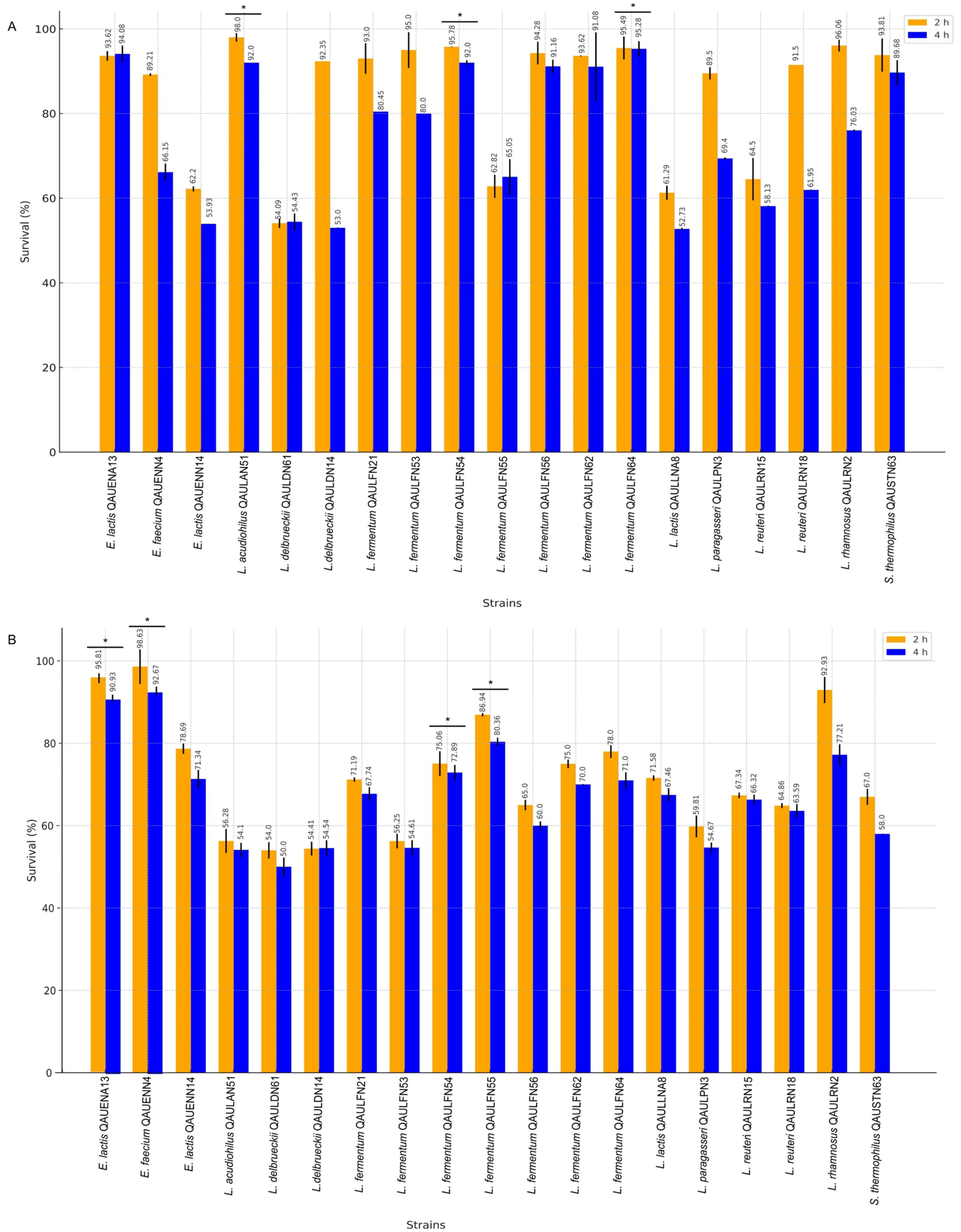


Fig. 1 Phylogenetic analysis and genomic comparisons of lactic acid bacteria

Survival under thein vitrosimulated fluids

The survival of the LAB isolates ranged from 55 to 90% after 4 h where *E. lactis* QAUEFNA13, *Lactobacillus paragasseri* QAULPN3 and *L. fermentum* QAULFN51 had statistically significant higher survival ($p < 0.05$) after 4 h of incubation (Fig. 3).


A
Survival (%)
Strains
2 h
4 h
0
20
40
60
80
100
93.62
94.08
89.21
66.15
62.2
53.93
98.0
92.0
54.09
54.43
92.35
53.0
93.0
80.45
95.0
80.0
95.78
92.0
62.82
65.05
94.28
91.16
93.62
91.08
95.49
95.28
61.29
52.73
89.5
69.4
64.5
58.13
91.5
61.95
96.06
76.03
93.81
89.68
*
E. lactis QAUENA13
E. faecium QAUENN4
E. lactis QAUENN14
L. acudiohilus QAULAN51
L. delbrueckii QAULDN61
L.delbrueckii QAULDN14
L. fermentum QAULFN21
L. fermentum QAULFN53
L. fermentum QAULFN54
L. fermentum QAULFN55
L. fermentum QAULFN56
L. fermentum QAULFN62
L. fermentum QAULFN64
L. lactis QAULLNA8
L. paragasseri QAULPN3
L. reuteri QAULRN15
L. reuteri QAULRN18
L. rhamnosus QAULRN2
S. thermophilus QAUSTN63
B
Survival (%)
Strains
2 h
4 h
0
20
40
60
80
100
95.81
90.93
98.63
92.67
78.69
71.34
56.28
54.1
54.0
50.0
54.41
54.54
71.19
67.74
56.25
54.61
75.06
72.89
86.94
80.36
65.0
60.0
75.0
70.0
78.0
71.0
71.58
67.46
59.81
54.67
67.34
66.32
64.86
63.59
92.93
77.21
67.0
58.0
*
E. lactis QAUENA13
E. faecium QAUENN4
E. lactis QAUENN14
L. acudiohilus QAULAN51
L. delbrueckii QAULDN61
L.delbrueckii QAULDN14
L. fermentum QAULFN21
L. fermentum QAULFN53
L. fermentum QAULFN54
L. fermentum QAULFN55
L. fermentum QAULFN56
L. fermentum QAULFN62
L. fermentum QAULFN64
L. lactis QAULLNA8
L. paragasseri QAULPN3
L. reuteri QAULRN15
L. reuteri QAULRN18
L. rhamnosus QAULRN2
S. thermophilus QAUSTN63

◂**Fig. 2** Percentage survival of LAB strains isolated from dahi samples at low pH and high bile salts. **A** Percentage survival of LAB at pH 2 after 2 and 4 h at 37 °C. **B** Percentage survival of LAB at 0.45% bile salt after 2 and 4 h at 37 °C. Strains having statistically significant survival ($p < 0.05$) are presented by asterisk sign 7

Antimicrobial activity of LAB isolates

Inhibitory effect of LAB against *B. subtilis* ATCC 19659 was observed highest as 85% and lowest as 66%. The growth of *B. subtilis* ATCC 19659 was significantly inhibited ($p < 0.05$) by *Lactobacillus delbrueckii* QAULDN61 (85.44%), *L. fermentum* QAULFN53 (82.76%), and *L. delbrueckii* QAULDN14 (82.27%) (Table 1). The inhibitory effect against *E. coli* ATCC 25922 was observed from 60.72 to 83.14%. The strains *E. lactis* QAUEFNA13, *L. acidophilus* QAULAN51, and *Lacticaseibacillus rhamnosus* QAULRN2 significantly inhibited ($p < 0.05$) the growth of *E. coli* ATCC 25922 with percentages of 83.14%, 83.14%, and 82.95%, respectively (Table 1). The isolates effected the growth of *P. aeruginosa* ATCC 15422 from a range from 52.43 to 88.96%. Statistically significant activity ($p < 0.05$) against *P. aeruginosa* ATCC 15422 was observed by the strains of *L. fermentum* QAULFN4 (88.96%), *L. acidophilus* QAULAN51 (88.07%), *L. fermentum* QAULFN56 (87.11%), and *L. delbrueckii* QAULDN14 (86.01%) (Table 1). The LAB isolates effected the growth of *S. enterica* ATCC 27870 in a range from 52.39 to 80.59%. The strains of *L. delbrueckii* QAULDN14 and *Streptococcus thermophilus* QAUSTN63 significantly inhibited ($p < 0.05$) the growth of *S. enterica* ATCC 27870 up to ranges of 80.59%, 80.03%, and 79.89%, respectively (Table 1). The growth of *S. aureus* ATCC 6538 was inhibited from 51.63 to 88.26%. The strains *L. fermentum* QAULFN56 significantly inhibited ($p < 0.05$) the growth of *S. aureus* (Table 1). The growth of *S. pneumoniae* ATCC 49619 was inhibited from 51.88 to 88.93%. The strains *L. acidophilus* QAULAN51 and *L. fermentum* QAULFN55 had significant inhibitory activity ($p < 0.05$) against *S. pneumoniae* ATCC 49619 (Table 1). Interestingly, MSPC inhibit all the pathogens at significant level. The inhibitory activity of the n-CFS was observed minimal toward the tested microorganisms; however, n-CFS from MSPC showed comparatively high activity (Table S3).

Antibiotic susceptibility

The LAB isolates were checked for susceptibility toward various antibiotics purchased from Sigma-Aldrich. The results were compared to the MIC cut off values recommended by EFSA. The strains *Lactococcus lactis* QAULLNA8 was found resistant to Tetracycline and Chloramphenicol. *L. rhamnosus* QAULRN2 was found resistant to ampicillin. Remaining strains were sensitive to the EFSA recommended antibiotics (Table S4).

Cell surface hydrophobicity and auto-aggregation

The isolates presented diverse CSH like *E. lactis* QAUELNN14 showed minimum (52%) while *E. lactis* QAUEFNA13 showed maximum (98%) hydrophobicity (Fig. 4). The isolates were checked for auto-aggregation ability that ranged from 52% for *E. lactis* QAUELNN14 to 93% for *L. fermentum* QAULFN53 (Fig. 4). The strains *L. lactis* QAULLNA8, *E. lactis* QAUELNA13, *L. reuteri* QAULRN15, *L. fermentum* QAULFN53, and *S. thermophilus* QAUSTN63 in comparison had significantly ($p < 0.05$) high CSH and auto-aggregation (Fig. 4). Similarly, MSPC also presented significantly ($p < 0.05$) high CSH and auto-aggregation (Fig. 4).

Exopolysaccharide production

All the strains were found EPS producing except the strains of *E. lactis* QAUEFNA13, *E. lactis* QAUELNN14, and *E. faecium* QAUEFNN2 (Table 2).

Bile salt hydrolase activity

Bile salt hydrolase activity of the stains helps determine their role in lipid metabolism and in the generation of short-chain fatty acids. The strains were evaluated for their BSH activity. Some strains showed BSH activity, while some did not present any activity (Table 2).

Thein vitrocholesterol removal

The management of body weight, obesity, and related diseases could benefit from LABs' ability to remove cholesterol in an in vitro environment. The isolates demonstrated excellent cholesterol-assimilating capacity, with individual and consortium efficiencies ranging from 34 to 98% (Fig. 5).

Decarboxylase activity

The strains were found negative for decarboxylase activities.

Antioxidant activity of the LAB isolates

L. fermentum QAULFN21 presented maximum (76%) and *L. lactis* QAULLNA8 presented minimum (37%) antioxidant activity. However, *L. paragasseri* QAULPN3, *L. delbrueckii* QAULDN14, *L. fermentum* QAULFN53, *L. fermentum* QAULFN54, *L. fermentum* QAULFN62, and MSPC were significantly ($p < 0.05$) high DPPH radical scavengers (Fig. 6).

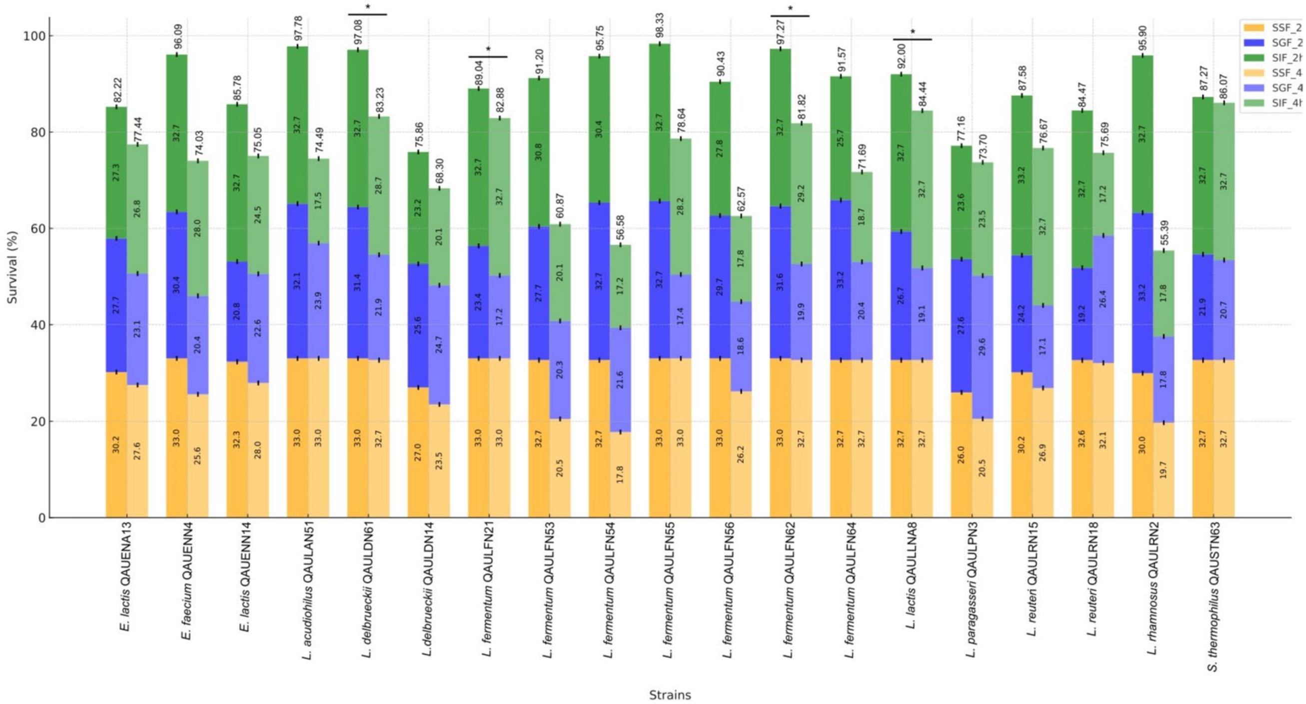


Fig. 3 Percentage survival of LAB isolated from dahi samples in the in vitro simulated gastrointestinal fluids after 2 and 4 h at 37 °C. Strains having statistically significant survival ($p<0.05$) are presented by asterisk sign

Table 1 Percentage inhibition of indicator bacterial pathogens by LAB strains isolated from traditional fermented milk product dahi after 24 h

LAB strains	*B. subtilis* ATCC 19659	*E. coli* ATCC 25922	*P. aeruginosa* ATCC 15422	*S. enterica* ATCC 27870	*S. aureus* ATCC 6538	*S. pneumonia* ATCC 49619
L. lactis QAULLNA8	71.17±1.17	60.72±0.72	69.97±0.82	79.47±0.38	59.64±0.51	51.88±0.46
E. lactis QAUEFNA13	66.32±1.32	**83.14±0.31**	56.26±1.19	52.39±0.89	55.64±1.57	75.76±0.23
E. faecium QAUEFNN4	79±1	81.10±0.34	71.75±0.77	77.52±0.42	67.66±1.41	73.83±0.25
E. lactis QAUELNN14	72.86±0.86	60.54±0.73	65.86±0.93	63.55±0.68	57.36±1.54	63.29±0.35
L. rhamnosus QAULRN2	79.29±1.29	**82.95±0.31**	55.72±1.21	75.00±0.47	51.63±1.62	65.75±0.33
L. paragasseri QAULPN3	71.87±0.87	70.54±0.54	80.53±0.53	71.93±0.52	78.25±1.28	71.72±0.27
L. delbrueckii QAULDN14	**82.27±0.27**	72.58±0.50	**86.01±0.38**	**80.03±0.38**	69.66±1.39	70.95±0.95
L. reuteri QAULRN15	81.48±0.48	62.76±0.68	82.86±0.46	67.60±0.61	63.37±1.47	73.83±0.25
L. reuteri QAULFN18	78.50±0.50	71.65±0.52	74.50±0.69	71.37±0.52	70.52±1.37	76.81±0.22
L. fermentum QAULFN21	73.55±0.55	70.54±0.54	72.44±0.75	69.42±0.57	79.10±1.26	84.01±0.15
L. acidophilus QAULAN51	80.58±0.58	**83.14±0.31**	**88.07±0.32**	77.24±0.42	74.53±1.32	**88.23±0.11**
L. fermentum QAULFN53	**82.76±0.76**	77.58±0.41	80.67±0.52	79.19±0.39	82.25±1.22	80.33±0.18
L. fermentum QAULFN54	76.72±0.72	68.13±0.59	84.50±0.42	73.89±0.49	80.25±1.25	83.84±0.15
L. fermentum QAULFN55	74.84±0.84	71.65±0.52	83.68±0.44	76.12±0.44	**88.26±1.15**	**85.24±0.14**
L. fermentum QAULFN56	81.57±0.57	72.02±0.51	**87.11±0.35**	71.65±0.53	80.25±1.23	70.67±0.28
L. delbrueckii QAULDN61	**85.44±0.44**	66.46±0.99	77.65±0.61	76.40±0.44	60.50±1.69	72.25±0.26
L. fermentum QAULFN62	70.08±1.08	78.32±0.40	82.86±0.46	79.75±0.38	81.39±1.23	82.26±0.17
S. thermophilus QAUSTN63	66.92±0.92	60.91±0.72	73.81±0.71	**79.89±0.37**	56.50±1.56	65.05±0.33
L. fermentum QAULFN64	81.38±0.38	61.09±0.72	**88.96±0.30**	75.42±0.46	82.25±1.22	81.73±0.18
MSPC	**84.93±0.76**	**82.01±0.44**	**89.00±0.51**	**80±1.12**	**86.00±0.15**	**88.93±1.00**

Values in bold are presenting significant values with $p\leq 0.05$

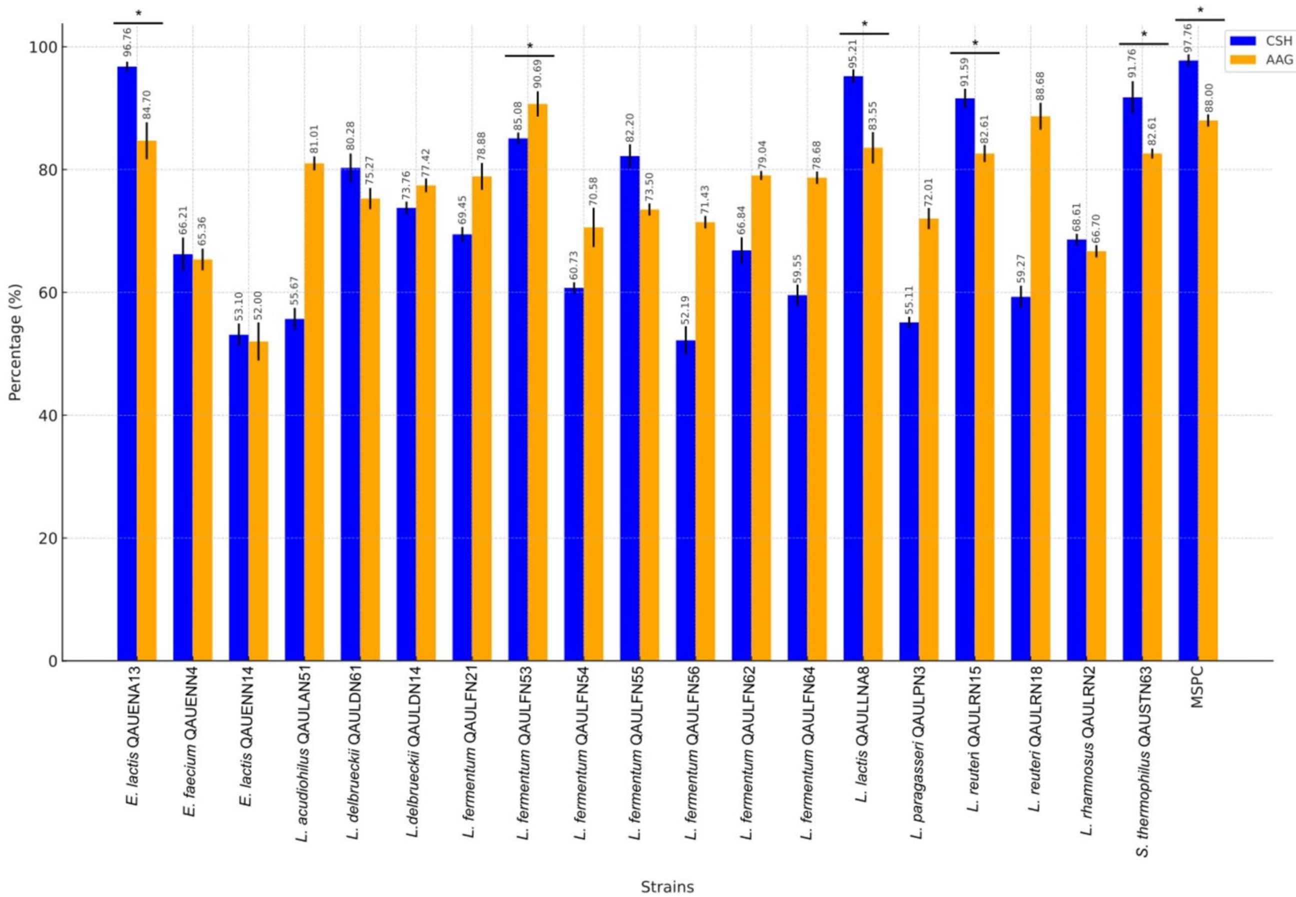


Fig. 4 Percentage cell surface hydrophobicity (CSH) and auto-aggregating (AAG) ability of lactic acid bacteria strains isolated from dahi samples. Strains having statistically significant activity ($p<0.05$) are presented by asterisk sign

Safety evaluation of isolates

The safety of the isolates was determined by their hemolytic and DNase activities, which proves their nonpathogenic and nontoxigenic statuses (Rychen et al. 2018). The results revealed no hemolytic and no DNase activity of the isolates.

Technological characterization of LAB isolates

The enzymatic potential of LAB strains

Among the strains, 12 were found proteolytic, 3 lipolytic, 9 amylolytic, and 15 cellulolytic (Table 2).

Survival of LAB isolates at a different salt concentration

L. lactis QAULLNA8 and *E. lactis* QAUEFNA13 had significantly ($p<0.05$) high survival in 4% NaCl and *L. reuteri* QAULRN15 and *L. fermentum* QAULFN54 had significantly ($p<0.05$) high survival in 6% NaCl (Fig. 7).

Growth of LAB isolates at different temperatures

The isolates were evaluated for growth at low (15 °C) and high (45 °C) temperatures. Each strain's growth pattern

Table 2 Enzymatic and functional profiles of the LAB strains isolated from dahi samples

LAB isolates	Enzymatic activity				Acid production	Milk curdling	BSH	EPS
	Proteolytic	Lipolytic	Amylolytic	Cellulolytic				
L. lactis QAULLNA8	+	-	+	-	+	+	-	+
E. lactis QAUEFNA13	+	+	+	-	+	+	-	-
E. faecium QAUEFNN4	-	-	-	-	-	+	-	+
E. lactis QAUELNN14	+	-	-	-	-	+	+	-
L. rhamnosus QAULRN2	-	-	+	+	+	+	+	+
L. paragasseri QAULPN3	+	+	-	+	+	+	+	+
L. delbrueckii QAULDN14	+	-	-	+	+	+	+	+
L. reuteri QAULRN15	+	-	-	+	+	+	+	+
L. reuteri QAULRN18	-	-	-	+	+	+	+	+
L. fermentum QAULFN21	-	+	+	+	+	+	+	+
L. acidophilus QAULAN51	+	-	+	+	+	+	+	+
L. fermentum QAULFN53	+	-	-	+	+	+	+	+
L. fermentum QAULFN54	+	-	-	+	+	+	+	+
L. fermentum QAULFN55	+	-	-	+	+	+	+	+
L. fermentum QAULFN56	-	-	-	+	+	+	+	+
L. delbrueckii QAULDN61	-	-	+	+	+	+	+	+
L. fermentum QAULFN62	-	-	+	+	+	+	+	+
S. thermophilus QAUSTN63	+	-	+	+	+	+	+	+
L. fermentum QAULFN64	+	-	+	+	+	+	+	+

+positive, −negative

was different, but all the strains are surviving at the tested temperatures. It was further observed that there was no statistically significant difference in the growth of any of the strains compared to the others, regardless of the temperature (Fig. 8).

General genomic features

Draft genomes of the strains ranged from 18,31,375 to 30,58,690 pb in case with GC content between 35 and 52.8%. The genomes had coding genes between 1877 and 3096 that were divided into subsystems ranging from 185 to 242. Detailed results are presented in Table 3.

Probiotic-related genes

The strains showed acid resistance through the presence of ATP synthase and ClpX ATP-binding subunits, with several strains also expressing Na^+/H^+ antiporters (Table 4). Genes encoding glucosamine-6-phosphate deaminase are uniformly present, indicating bile resistance strains, while genes for cholylglycine hydrolase is absent in *L. rhamnosus* QAULRN2 and *L. acidophilus* QAULAN51, suggesting variability in bile salt hydrolysis capacity (Table 4). Adhesion-related genes, such as glycosyltransferase and polysaccharide biosynthesis proteins, are widely present in the strains. Similarly, oxidative stress resistance was supported by genes like ATP-dependent Clp endopeptidase and thioredoxin-disulfide reductase, though genes for enzymes like glutathione peroxidase and manganese catalase were absent in some strains (Table 4). Temperature and osmotic resistance were consistently supported by the presence of genes for chaperonins GroEL, GroES, and ABC transporters (Table 4). Detailed results are presented in supplementary materials (supplementary file 2).

Mobile genetic element analysis

The genomes of the isolates were mined for the presence of the prophages. It detected intact, incomplete, or questionable prophages. The *E. faecium* QAUEFNN4 had 11 prophages, of which 3 were intact, 4 were incomplete, and 4 were questionable. The *L. reuteri* QAULRN15 had 9 prophages, 3 were intact, 5 were incomplete, while only 1 is questionable. The *L. fermentum* QAULFN51 had only one incomplete phage. The *L. fermentum* QAULFN53 had only one intact phage (Table S5). The genomes were mined for the presence of CRISPR sequences. Except *E. lactis* QAUELNN14, *L. fermentum* QAUEFN54, and *S. thermophilus* QAUSTN63, all the genomes had CRISPR sequences (Table S6).

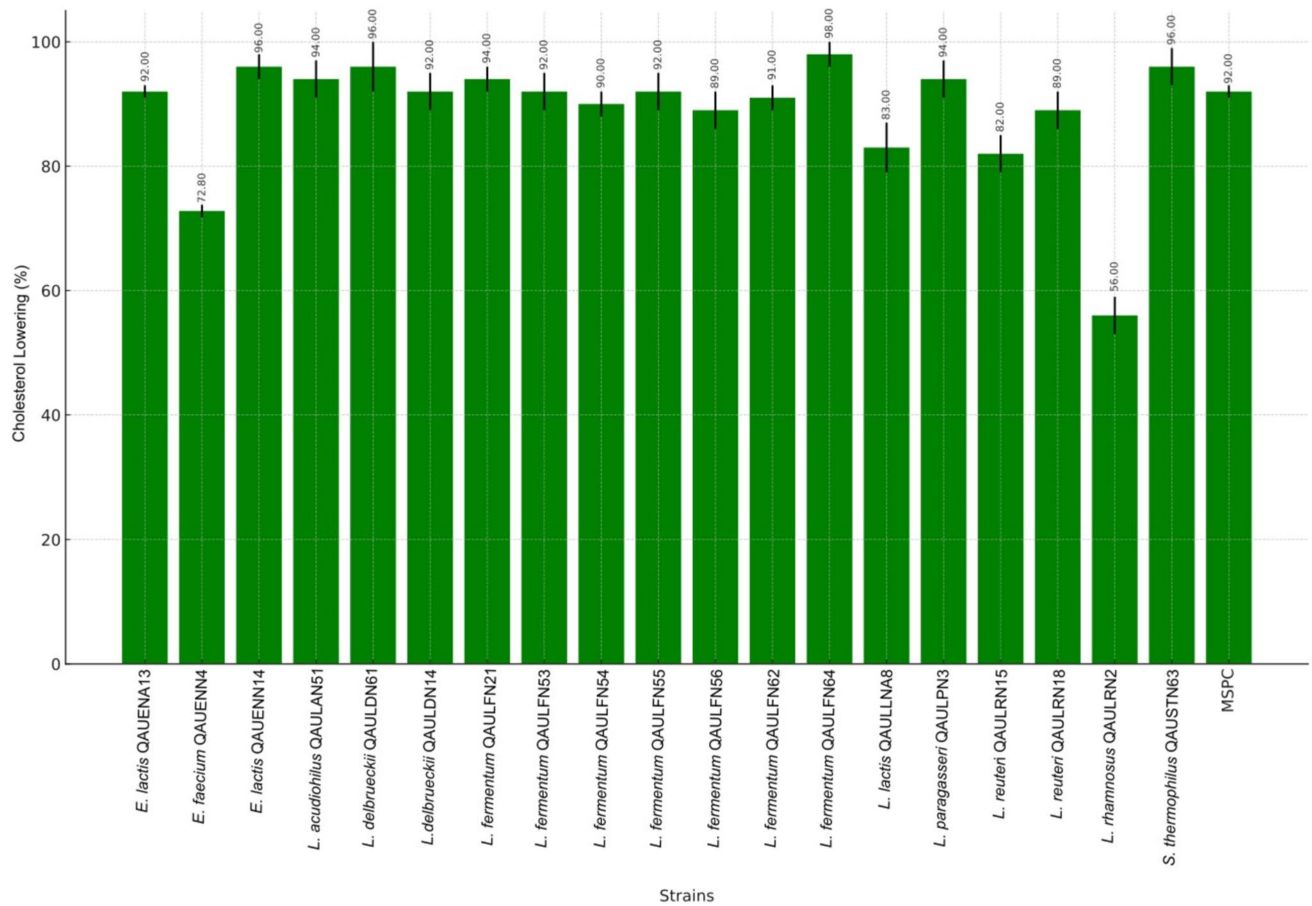


Fig. 5 Percentage cholesterol removal by LAB isolated from dahi samples after 24 h at 37 °C

Bacteriocin and secondary metabolic product

It was observed that Enterolysin A is widespread, found in strains *E. lactis* QAUEFNA13, *E. faecium* QAUEFNN4, and multiple *L. fermentum* strains (Table S7). *L. acidophilus* QAULAN51 exhibits both Acidocin J1132 β peptide and Bacteriocin helveticin J. *S. thermophilus* QAUSTN63 harbor Streptide and sactipeptides (Table S7).

The genomes were analyzed for secondary metabolic clusters. It was observed that *E. faecium* QAUEFNN4 has a RiPP cluster identical to Enterocin A (100%), while *L. lactis* QAULLNA8 and *L. acidophilus* QAULAN51 feature RiPP-like clusters, showing 33% similarity to Gassericin T. *L. paragasseri* QAULPN3 contains RiPP-like clusters with similarities to Gassericin-S (66%) and Gassericin T (55%). *S. thermophilus* QAUSTN63 has RaS-RiPP clusters, including a Streptide match (100%). Detailed results are presented in Table S8.

Carbohydrate active enzymes

The strains carried different CAZymes, including glycoside hydrolases (GHs), glycosyl transferases (GTs), carbohydrate esterases (CEs), and carbohydrate-binding modules (CBMs). *E. faecium* QAUEFNN4 exhibits significant activity in GT1 (10) and GT2 (4), while *E. lactis* QAUEFNA13 and *E. lactis* QAUELNN14 show moderate activity across GH13 and CBM50, with 5 and 2 entries, respectively (Fig. 9). *L. acidophilus* QAULAN51 demonstrates considerable enzymatic activity in GT1 (8) and GT2 (3), while *L. delbrueckii* QAULDN14 and QAULDN61 have similar profiles, particularly in GH13 and CBM48 (Fig. 9). *L. fermentum* strains display consistent enzymatic activity across multiple enzymes, particularly CBM50, GH13, and GT4, while *L. lactis* QAULLNA8 and *L. paragasseri* QAULPN3 show diverse enzymatic activity in GHs and GTs (Fig. 9). The genomes of LAB demonstrate their ability to metabolize

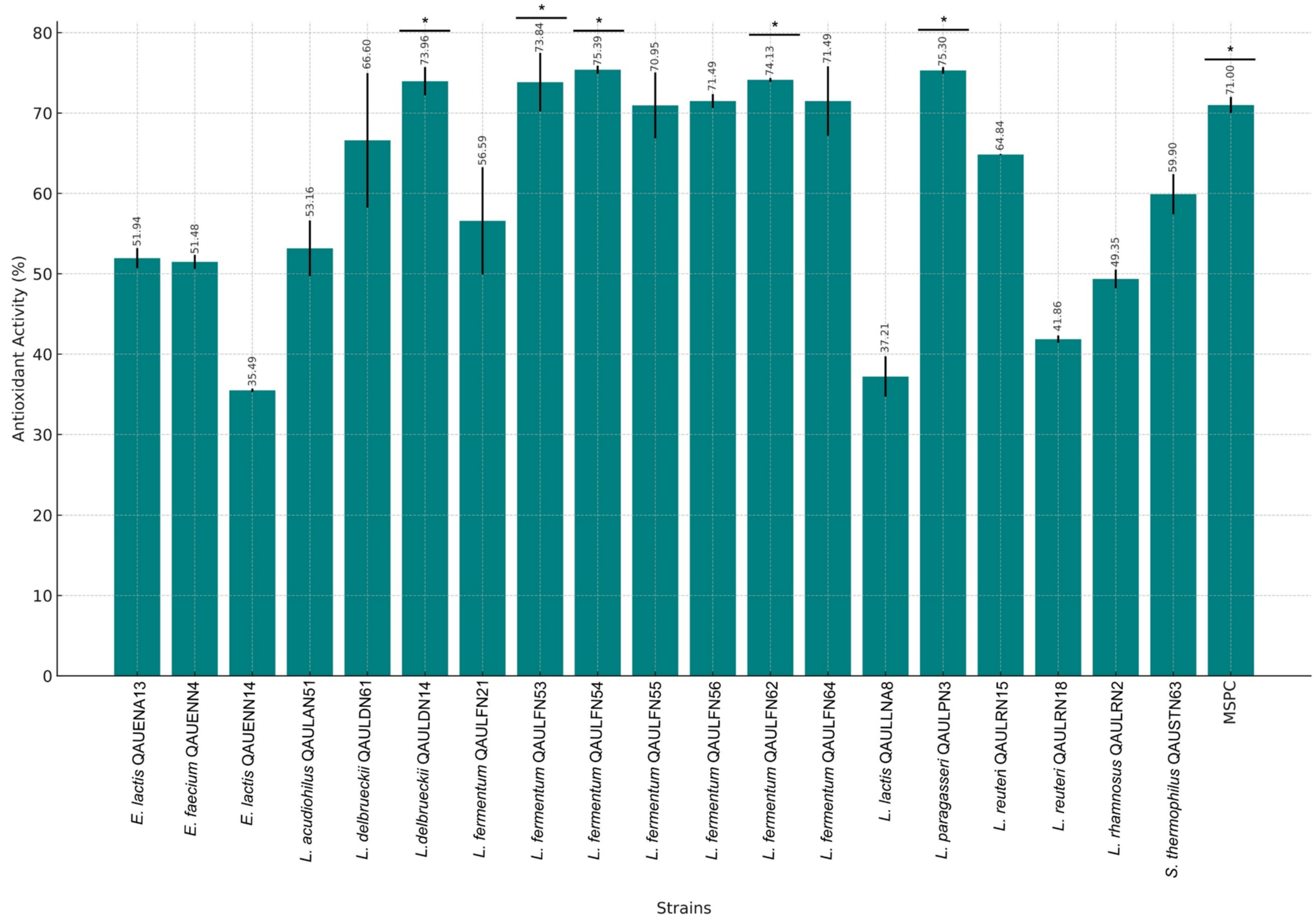


Fig. 6 Percentage antioxidant activity by LAB strains isolated from dahi samples. Strains with statistically significant activity ($p < 0.05$) are presented by asterisk sign

various carbohydrates (e.g., starch, cellulose, pectin, xyloglucan, xylan, fructan, and host-driven glycans) into monosaccharides and subsequently produce SCFAs, such as acetate, lactate, propionate, succinate, and butanoate. The metabolic pathway involves key intermediates like pyruvate, acetyl-CoA, and lactaldehyde, highlighting the diverse metabolic potential of LAB in polysaccharide utilization and SCFA biosynthesis (Supplementary Material 2).

Safety of the strains

The genomes of the lactic acid bacteria were found to contain no antibiotic resistance genes, no virulence factors, and no pathogenicity-associated islands. Furthermore, the probability of them being human pathogens was less than 0.9 (Table S9).

Survival of MSPC in TSI

During fasting state, before introducing to the TSI model, *Lactobacillus* had mean ASVs of 20,000/mL, which remained stable at 20,440 ASVs/mL by the end. *Streptococcus* had 17 ASVs/mL that reached to 440 ASVs/mL. *Lactococcus* had 407 ASVs/mL that reached to 1400 ASVs/mL. *Enterococcus* had 1537 ASVs/mL that reached to 3620 ASVs/mL (Fig. 10A).

During feeding state, before introducing to the TSI model, *Lactobacillus* had mean ASVs of 576/mL that reached to 33,280 ASVs/mL at the end. *Streptococcus* had 354 ASVs/mL that reached to 1880 ASVs/mL. *Lactococcus* had 69 ASVs/mL that reached to 416 ASVs/mL. *Enterococcus* had mean 402 ASVs/mL that reached to 17,090 ASVs/ml (Fig. 10B).

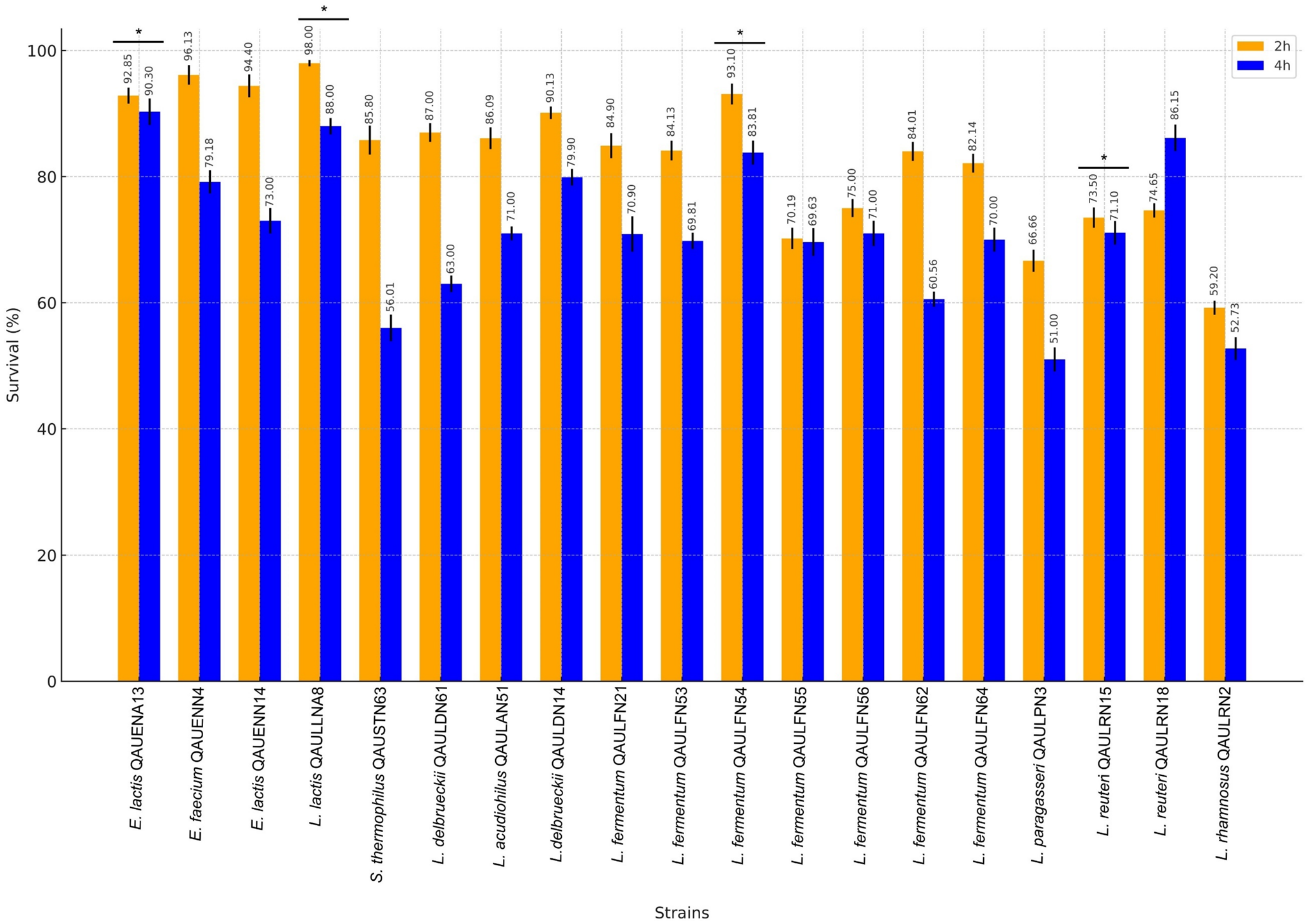


Fig. 7 Percentage survival of lactic acid bacteria strains isolated from dahi samples in 4% and 6% NaCl concentrations after 2 and 4 h of incubation at 37 °C. Coccus strains are evaluated for their survival in 4% NaCl and *Bacillus* strains are evaluated for their survival in 6% NaCl. Strains having statistically significant survival ($p<0.05$) are presented by asterisk sign

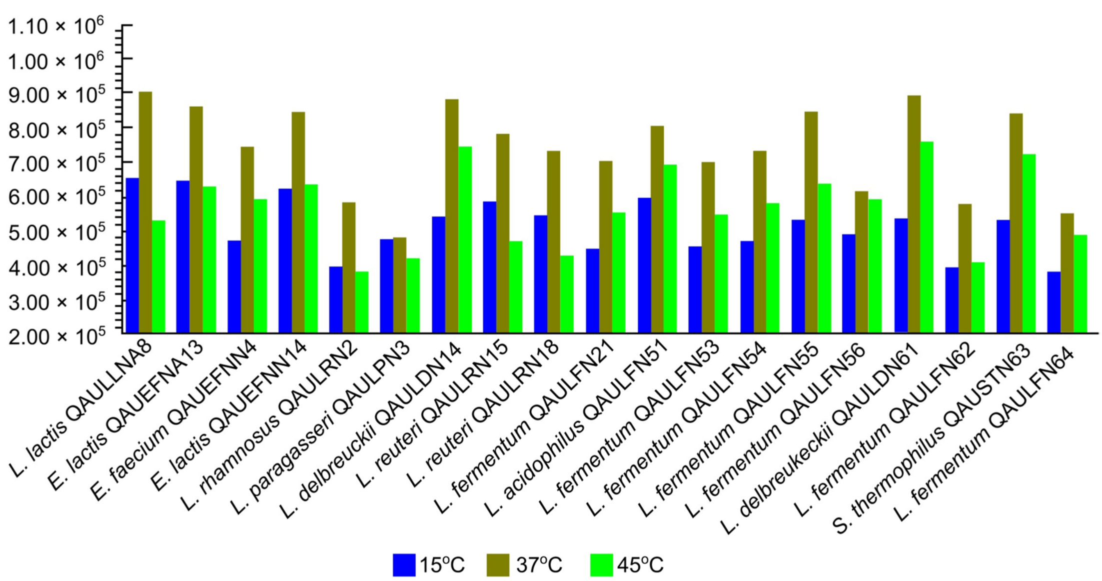


Fig. 8 Growth of lactic acid bacteria strains isolated from dahi samples at different temperatures (15 °C, 37 °C, and 45 °C) after 24 h

Table 3 Features of draft genomes from lactic acid bacteria strains isolated from traditionally fermented milk product dahi

LAB strains	Genome statistics							
	Size (bp)	GC (%)	N50(bp)	L50(count)	Contigs	Coding genes	COGs	Subsystems
E. faecium QAUEFNN4	30,58,690	38	2,03,093	5	4	3096	2518	229
E. lactis QAUEFNA13	26,72,289	38.2	1,00,275	9	81	2692	2292	224
E. lactis QAUELNN14	26,25,750	38.3	84,626	12	61	2626	2187	227
L. acidophilus QAULAN51	19,53,269	34.6	1,67,684	3	21	1919	1633	186
L. delbrueckii QAULDN14	19,94,221	49.3	26,573	20	155	2176	1706	194
L. delbrueckii QAULDN61	18,31,175	49.8	1,07,421	4	44	2022	1611	187
L. fermentum QAULFN21	18,67,005	52.8	37,092	15	74	1883	1595	212
L. fermentum QAULFN53	20,77,539	51.7	37,704	16	170	2177	1800	220
L. fermentum QAULFN54	19,66,551	52	7,651	75	129	2121	1749	215
L. fermentum QAULFN55	20,11,311	51.8	24,426	24	137	2085	1647	220
L. fermentum QAULFN56	19,00,053	52.4	29,869	19	102	1929	1715	207
L. fermentum QAULFN62	19,68,193	52	55,235	12	80	2004	1653	213
L. fermentum QAULFN64	19,00,053	52.4	29,869	19	102	1929	1684	207
L. lactis QAULLNA8	25,53,575	35	51,270	14	151	2716	2093	242
L. paragasseri QAULPN3	19,32,640	34.7	1,08,411	7	38	1877	1634	185
L. reuteri QAULRN15	21,51,450	38.6	13,579	52	391	2410	1882	217
L. reuteri QAULRN18	22,73,479	38.2	34,649	24	191	2384	1,730	219
L. rhamnosus QAULRN2	29,18,787	46.7	2,17,890	4	138	3001	2333	237
S. thermophilus QAUSTN63	18,38,081	38.8	1,23,061	5	46	2106	1703	214

Discussion

It is widely accepted in Pakistan and neighbouring countries that dahi possesses therapeutic characteristics, which can be directly or indirectly linked to the indigenous microorganisms present in the dahi (Akhtar et al. 2016; Burke et al. 2018). Bacteria originating from dahi could be used as probiotics for therapeutic purposes, for beneficial modulation of gut microbiota and in the formulation of functional foods and feeds. This study emphasizes the probiotic potential of LAB, both as individual strains and as a consortium, isolated from dahi.

The production of acids, utilization of lactose, and curdling of milk provided phenotypic confirmation of the isolates as LAB (Bozoudi et al. 2015; Mora et al. 2018). This identification was further confirmed through more accurate and reliable molecular techniques like ON-Rep-Seq, ANI, and GDGC (Krych et al. 2019; Li et al. 2024). With respect to fermentation mode, *L. fermentum* is typically considered heterofermentative but several strains in the current study exhibited homofermentative behavior. Certain strains of *L. fermentum* are reported to be homofermentative (Basso et al. 2014; Wafula et al. 2023). The ability of the strains to utilize different sugars is consistent with previously published results, which have shown that LAB can utilize different sugars as sole carbon source (Wang et al. 2021).

Resistance to pH, bile salts, NaCl and temperatures are the prerequisite for probiotics to survive in the host's intestine and food metrics (Ho & Yin Sze 2018). Probiotic is capable of conferring its associated beneficial effects to the host as long as at least 50% of the total intake remains viable in the host (Hill et al. 2014). The tested strains showed a survival rate over 50% in low pH and bile salts, with significant resilience after 4 h of exposure. These findings are consistent with previous studies that reported that dairy-origin LAB can survive in acidic pH and up to 0.5% bile salts (Haghshenas et al. 2017; Zoumpopoulou et al. 2018). The strains showed similar results in the simulated intestinal fluids and demonstrated significant resilience in the simulated intestinal model (TSI), even when in a consortium. These conditions mimic the actual gut environment and further validate that the strains are more likely to survive in the human intestinal tract (Cieplak et al. 2018; Queiroz et al. 2022). These findings are also evidenced at the genomic level, where most of the strains carried genes for ATPase, Na^+/H^+ antiporters, ATP synthase, glucosamine-6-phosphate deaminase, and choloylglycine hydrolase. Studies have reported the role of these genes in acid and bile tolerance in LAB (Kim et al. 2022).

LAB produce organic acids, antimicrobial proteins or peptides that act antagonistically to pathogenic microorganisms (Pisoschi et al. 2018). The LAB showed antagonistic activity against ATCC strains of *E. coli, S. aureus, P. aeruginosa, S. enterica, S. pneumoniae,* and *B. subtilis* at various rates ranging from 51 to 88%. However, when in consortium the effect was significant for each pathogen. Similar results

Table 4 Probiotic-related genes present in the genomes of lactic acid bacteria isolated from traditionally fermented milk product dahi

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
Acid resistance	Alkaline shock response	+	+	+	+	+	-	-	-	-	+
	ATP synthase	+	+	+	+	+	+	+	+	+	+
	ATP-dependent Clp protease	+	+	+	+	+	+	+	+	+	+
	GTP pyrophosphokinase protein	-	+	+	+	+	+	+	+	+	+
	Na^+/H^+ antiporter	+	+	+	+	-	+	+	+	+	+
	PTS system	+	+	+	+	+	+	+	+	+	+
	Pyruvate kinase	+	-	+	+	+	+	+	+	+	+
	Recombinase protein	-	+	+	-	-	+	+	+	+	+
	Response regulator	+	+	+	+	+	+	+	+	+	+
	Phosphoglycerate kinase	+	+	-	+	+	+	+	+	+	+
Bile resistance	Glucosamine-6-phosphate deaminase	+	+	+	+	+	+	+	+	+	-
	Oligopeptide ABC transporter substrate-binding protein	+	+	+	+	+	+	+	+	+	-
	Choloylglycine hydrolase	-	+	+	+	-	+	+	-	+	+
	CTP synthase	+	+	-	+	+	+	+	+	+	+
	Manganese-dependent inorganic pyrophosphatase	+	+	-	+	+	+	+	+	+	+

Table 4 (continued)

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
Adhesion	3-Oxoacyl-[acyl-carrier-protein] reductase	+	+	+	+	-	+	-	-	-	-
	Class A sortase	-	+	+	+	+	+	+	+	+	+
	Tyrosine-protein kinase	-	+	+	-	+	+	+	+	+	+
	Glycosyltransferase family 2 protein	+	+	+	+	+	+	+	+	+	+
	Polysaccharide biosynthesis protein	+	+	+	+	+	+	+	+	+	+
	Prepilin-type N-terminal cleavage/methylation domain-containing protein	+	+	+	+	-	+	+	+	+	+
	Sugar transferase	+	+	+	+	+	+	+	+	+	+
	UDP-glucose 4-epimerase GalE	+	+	+	+	+	+	+	+	+	+
Oxidative resistance	Alkyl hydroperoxide reductase	+	+	+	+	-	-	-	-	-	-
	Stress response protein	+	+	+	+	+	+	+	+	+	+
	ATP-dependent Clp endopeptidase	-	+	+	+	+	+	-	-	+	+
	Glutathione peroxidase	+	+	+	+	+	-	-	-	-	-
	Manganese catalase	-	+	+	+	-	-	-	-	-	-

Table 4 (continued)

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
	Molecular chaperone	+	+	+	+	+	+	+	+	+	+
	NADP-dependent oxidoreductase	+	+	+	+	+	-	-	-	-	-
	Organic hydroperoxide resistance protein	-	+	+	+	-	-	-	-	-	-
	Redox-sensing transcriptional repressor Rex	+	+	+	-	+	+	+	+	+	+
	SOS response-associated peptidase family protein	+	+	+	+	-	+	-	-	+	-
	Superoxide dismutase	+	+	+	+	-	-	-	-	-	-
	Thioredoxin-disulfide reductase	+	+	+	+	+	+	+	+	+	+
	Glyceraldehyde-3-phosphate dehydrogenase	+	+	+	+	+	+	+	+	+	+
	Universal stress protein	+	+	+	+	+	+	+	+	+	+
Temperature resistance	Chaperonin GroEL	+	+	+	-	+	+	+	+	+	+
	Co-chaperone GroES	+	+	+	+	+	+	+	+	+	+
	Cold-shock protein	+	+	+	+	+	-	+	+	-	+

Table 4 (continued)

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
Osmotic resistance	Choline family ABC transporter	-	-	+	+	+	-	-	-	-	-
	ABC transporter	+	+	+	+	+	+	+	+	+	+
Stress response	Product	L. fermentum QAULFN54	L. fermentum QAULFN21	L. fermentum QAULFN55	L. fermentum QAULFN56	L. fermentum QAULFN62	L. fermentum QAULFN64	L. reuteri QAULRN15	L. reuteri QAULRN18	S. thermophilus QAUSTN63	
Acid resistance	Alkaline shock response	+	+	+	+	+	+	+	+	+	
	ATP synthase	+	+	+	+	+	+	+	+	+	
	ATP-dependent Clp protease	+	+	+	+	+	+	+	+	+	
	GTP pyrophosphokinase protein	+	+	+	+	+	+	+	+	+	
	Na^+/H^+ antiporter	-	-	+	-	-	-	+	+	-	
	PTS system	+	+	+	+	+	+	+	-	+	
	Pyruvate kinase	+	+	+	+	+	+	+	+	+	
	Recombinase protein	+	-	+	-	+	-	+	+	-	
	Response regulator	+	+	+	+	+	+	+	+	+	
	Phosphoglycerate kinase	+	+	+	+	+	+	+	+	+	
Bile resistance	Glucosamine-6-phosphate deaminase	-	-	-	-	-	-	-	+	+	
	Oligopeptide ABC transporter substrate-binding protein	-	-	-	-	-	-	-	-	-	
	Choloylglycine hydrolase	+	+	+	+	+	+	+	+	-	
	CTP synthase	+	+	+	+	+	+	+	+	+	

Table 4 (continued)

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
	Manganese-dependent inorganic pyrophosphatase	+	+	+	+	+	+	+	+	+	
Adhesion	3-Oxoacyl-[acyl-carrier-protein] reductase	-	-	-	-	-	-	+	-	+	
	Class A sortase	+	+	+	+	+	+	+	+	-	
	Tyrosine-protein kinase	+	+	-	+	+	+	-	+	+	
	Glycosyltransferase family 2 protein	+	+	+	+	+	+	+	+	+	
	Polysaccharide biosynthesis protein	+	+	+	+	+	+	+	+	+	
	Prepilin-type N-terminal cleavage/methylation domain-containing protein	+	+	+	+	+	+	-	+	-	
	Sugar transferase	+	+	+	+	+	+	+	+	+	
	UDP-glucose 4-epimerase GalE	+	+	+	+	+	+	+	+	+	
Oxidative resistance	Alkyl hydroperoxide reductase	-	-	-	-	-	-	-	-	-	
	Stress response protein	+	+	+	+	+	+	+	+	+	
	ATP-dependent Clp endopeptidase	+	+	+	+	+	+	+	+	-	

Table 4 (continued)

Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAUEFNA13	*E. lactis* QAUELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. paragasseri* QAULPN3	*L. fermentum* QAULFN53
	Glutathione peroxidase	-	-	-	-	-	-	-	-	-	
	Manganese catalase	-	-	-	-	-	-	-	-	-	
	Molecular chaperone	+	+	+	+	+	+	+	+	+	
	NADP-dependent oxidoreductase	-	-	-	-	-	-	-	-	-	
	Organic hydroperoxide resistance protein	-	-	-	-	-	-	-	-	-	
	Redox-sensing transcriptional repressor Rex	+	+	+	+	+	+	+	+	+	
	SOS response-associated peptidase family protein	-	-	-	-	-	-	-	-	-	
	Superoxide dismutase	-	-	-	-	-	-	-	-	+	
	Thioredoxin-disulfide reductase	+	+	+	+	+	+	+	+	+	
	Glyceraldehyde-3-phosphate dehydrogenase	+	+	+	+	+	+	+	+	+	
	Universal stress protein	+	+	+	+	+	+	+	+	+	

Table 4 (continued)

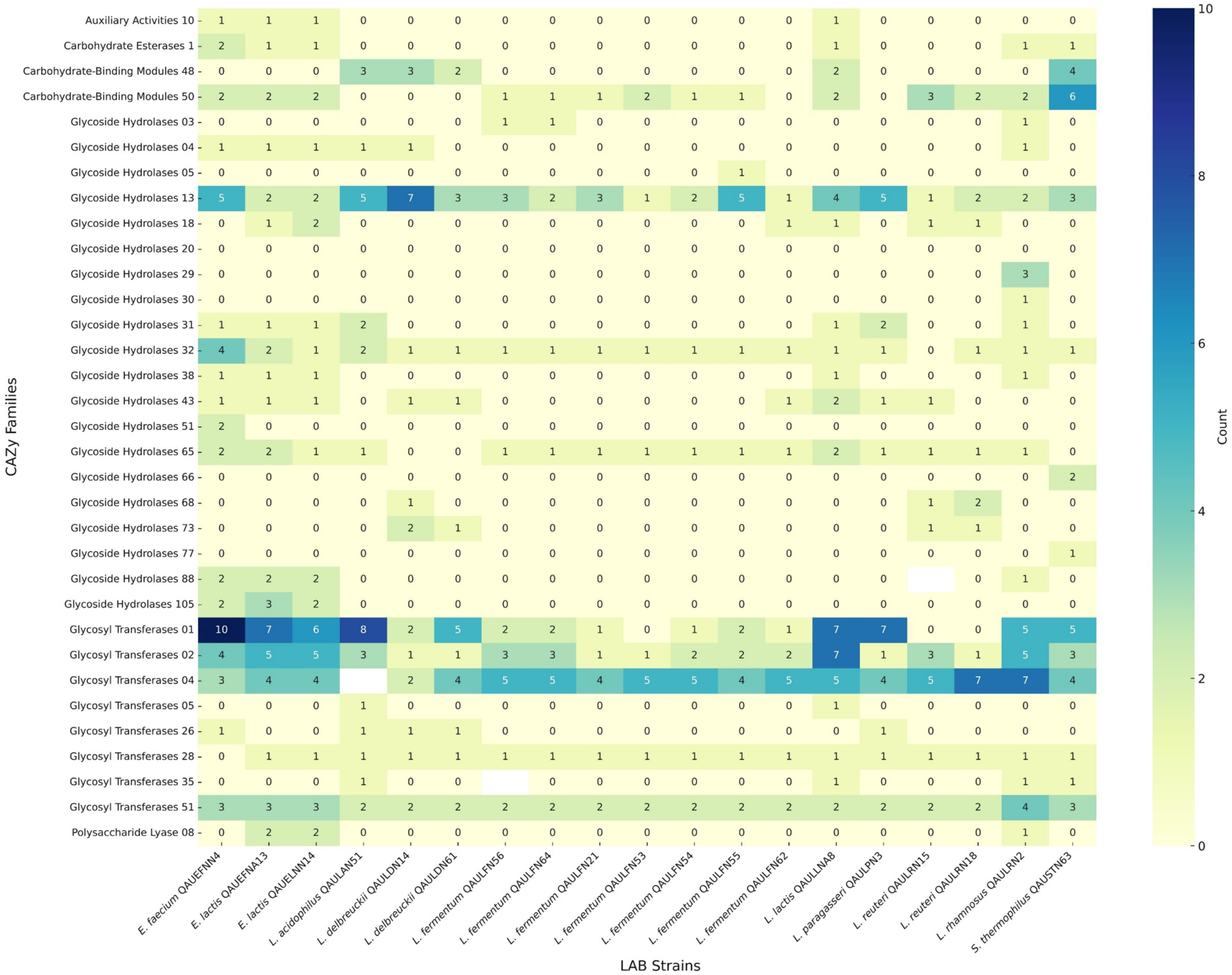
Stress response	Product	*L. lactis* QAULLNA8	*E. faecium* QAUEFNN4	*E. lactis* QAU-EFNA13	*E. lactis* QAU-ELNN14	*L. rhamnosus* QAULRN2	*L. acidophilus* QAULAN51	*L. delbrueckii* QAULDN14	*L. delbrueckii* QAULDN61	*L. para-gasseri* QAULPN3	*L. fermentum* QAULFN53
Temperature resistance	Chaperonin GroEL	+	+	+	+	+	+	+	+	+	
	Co-chaperone GroES	+	+	+	+	+	+	+	+	+	
	Cold-shock protein	+	+	+	+	+	+	-	+	+	
Osmotic resistance	Choline family ABC transporter	-	-	-	-	-	-	-	-	-	
	ABC transporter	+	+	+	+	+	+	+	+	+	

have been reported where LAB isolated from dairy products had antagonistic activity against foodborne pathogens (Leite et al. 2015). Research has indicated that LAB in consortia may serve as alternatives to antibiotics in both human health and animal production settings (Vieco-Saiz et al. 2019). Genomic analysis further confirmed phenotypic results by revealing genes responsible for producing bacteriocins especially Enterolysin A and Acidocin in various strains.

CSH and auto-aggregation are important for bacteria to attached and colonize the intestinal tract (Puniya et al. 2016). High (> 50%) CSH and auto-aggregation was observed in our strains that are in line with a recent study, reporting high hydrophobicity and auto-aggregation for LAB strains (Kathiriya et al. 2018). MSPC exhibited significant CSH and auto-aggregating activity. Combinations of strains with varied surface properties may result in better aggregation; some strains may form stronger biofilms or clusters when combined together with complementary strains (Joshi et al. 2021). These phenotypic properties are further validated by the genomic presence of genes involved in adhesion, such as glycosyltransferase and polysaccharide biosynthesis proteins, which enhance the binding ability of LAB to intestinal surfaces (Aziz et al. 2023). EPS produced by LAB can enhance biofilm formation and modulate the immune system (Fong et al. 2016). Most isolates in this study, except for *E. lactis* QAUEFNA13, were EPS producers, which aligns with previous research indicating that EPS production contributes to LAB's probiotic efficacy (Fong et al. 2016).

A factor of safety concern, probiotic must be sensitive to the antibiotics that have been recommended by EFSA (Álvarez-Cisneros & Ponce-Alquicira 2018). The isolates were found to be susceptible to the recommended antibiotics, similar to previously published results (Ren et al. 2018). Genomic mining also retuned no hit for antibiotic resistance genes in the genomes. The strains showed no hemolysis and DNase activity when evaluated on respective media. The probability of being a pathogen in the genomes was very low (< 1), and these strains were predicted to be non-human pathogens. The strains have no virulent or pathogenic genes and therefore are safe. Recently, it has reported that LAB isolated from dairy products have no hemolysis activity on blood agar and no DNase activity on DNase media (Zarour et al. 2018).

Some probiotics can remove cholesterol from the culture media, which suggests that when these microorganisms are consumed, they could perform a similar function in vivo thus helping to control plasma cholesterol. The strains were observed to have bile salt hydrolase and cholesterol-lowering effects, which is supported by previously published studies (Castorena-Alba et al. 2018; Singhal et al. 2012). These results are also supported at the genomic level, as the genes required for bile resistance and cholesterol assimilation are present in the genomes of the strains. Choloylglycine hydrolase genes, widely present in the strains, are reported



SiFig. 9 Heatmap for distribution of the number of genes of carbohydrate-active enzymes in lactic acid bacteria strains isolated from traditional fermented milk product dahi

to play a role in bile deconjugation and cholesterol removal in LAB (Frappier et al. 2022). Excessive accumulation of various reactive oxygen species (ROS) damages proteins, lipids, and DNA, leads to tissue damage and organ dysfunction (Halliwell & Gutteridge 1984). The strains of LAB exhibited varied antioxidant activities (37–76%) that is also supported by previous studies (Kim et al. 2020; Lin et al. 2018). For antioxidant activity, we used DPPH which has been used widely as a radical to evaluate antioxidant activity. However, DPPH is stable in methanol and is not a naturally occurring ROS, so the bacteria may exhibit different activity when assessed using a different method (Antolovich et al. 2002). These results were also confirmed through genomic analysis, where genes responsible for antioxidant activity, previously reported in the literature, were present in the genomes of the strains (Feng & Wang 2020). The higher cholesterol-lowering and antioxidant activities observed in LAB consortia compared to individual strains may be due to synergistic metabolic interactions and their diverse capabilities (Duncker et al. 2021).

Strains in the current study were observed to have proteolytic, lipolytic, amylolytic, and cellulolytic activities similar to previous findings (Hanchi et al. 2018). Proteolytic LAB produce bioactive peptides with antihypertensive, antioxidant, antimicrobial, and immunomodulatory effects (García-Cano et al. 2019). Strains with high lipolytic activity can be used as starter or adjunct cultures and can also be used for cholesterol assimilation activities in in vivo (Dora & Glenn 2002). It has been reported that the amylolytic bacteria can degrade the starch and convert it into low molecular weight sugars and have many industrial applications (Mehta & Satyanarayana 2016). Cellulolytic LAB can improve fiber digestion and release of beneficial compounds like phenolics (Feng & Wang 2020). Amylolytic LAB are naturally present in fermented

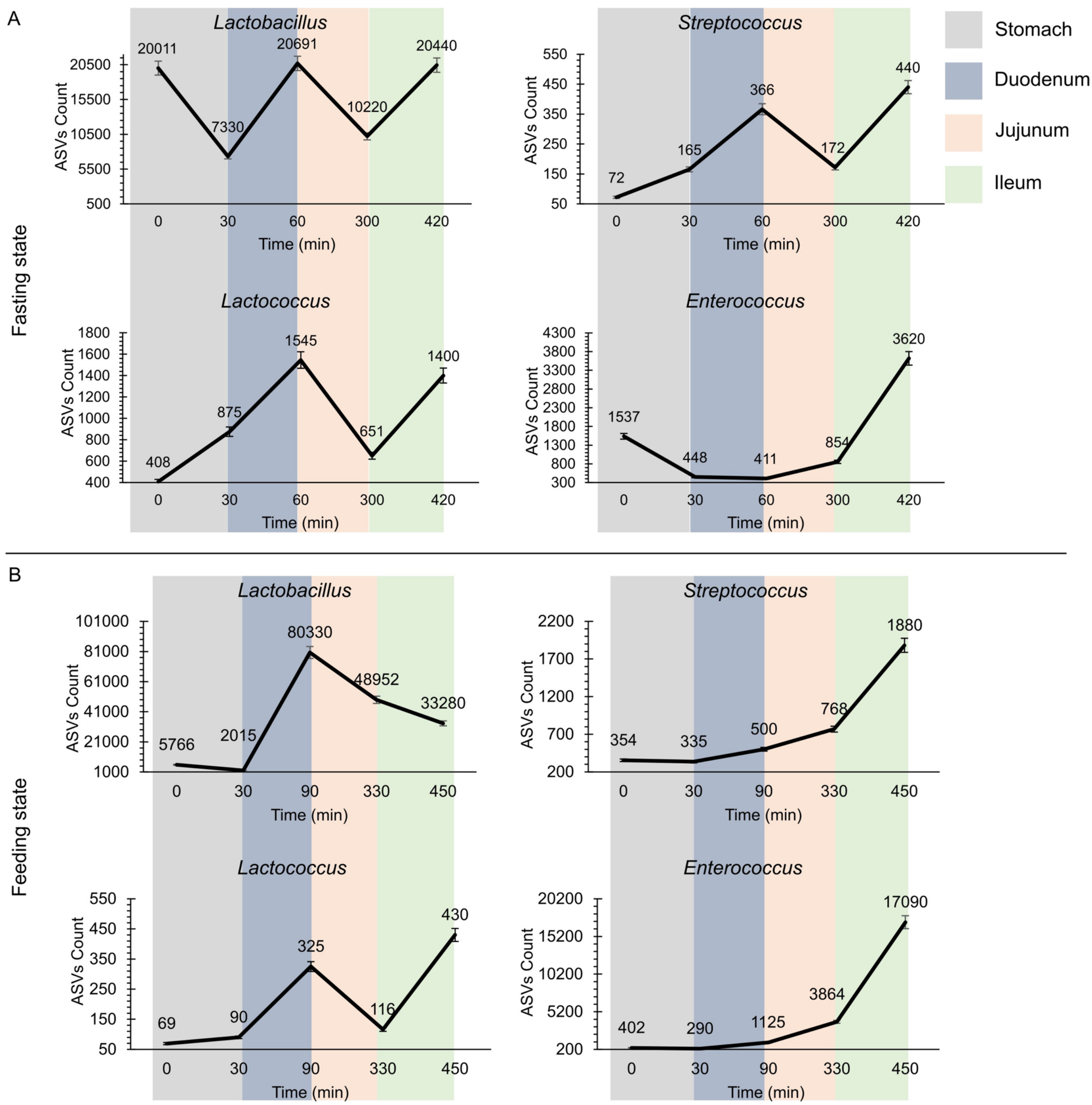


Fig. 10 Survival of MSPC in simulates small intestine (TSI). **A** 16S rDNA-based survival of MSPC in TSI during fasting phase. **B** 16S rDNA-based survival of MSPC in TSI during feeding phase

foods, plants matrixes, plant wastes, and the digestive tracts of humans and animals (Marco et al. 2017). Genomic analysis revealed the presence of carbohydrate-active enzymes such as glycoside hydrolases and glycosyltransferases, which are essential for these metabolic functions.

NaCl is a common food preservative that is used to extend the shelf life of many food products. The ability of LAB to survive in a high-salt environment is an important characteristic for ensuring the safety and quality of these products. The strains *L. lactis* QAULLNA8 and *E. lactis* QAUEFNA13 had significant survival in 4% NaCl while the strains *L. reuteri* QAULRN15 and *L. fermentum* QAULFN54 had significant survival in 6% NaCl. These reports are in line with recent published studies on the survival of LAB in different concentrations of NaCl (Reale et al. 2020). When compared to the growth at ambient temperature (37 °C), it was observed that the isolates are able to grow at 15 °C and 45 °C without any significant difference.

These findings are supported by previous study where LAB isolated from milk were able to grow at minimum of 5 °C to maximum of 55 °C (Ahmed et al. 2006). Genomic findings such as GroEL and GroES chaperonins as well as cold shock proteins could explain their ability to endure extreme environments. Though we conducted a detail characterization of the LAB isolated from dahi, these are limited to the in vitro trials, which could be considered the main limitation of the current study. Though the strains showed synergy when grown in consortium, a more detailed study is needed at both the laboratory and genomic levels. Further in vivo research is necessary to fully assess their efficacy and therapeutic potential in human applications.

Conclusion

The phenotypic and genotypic characterization of LAB presented strong probiotic potential, with resilience in gastrointestinal conditions, robust antimicrobial properties, and significant health benefits like cholesterol reduction and antioxidant activity. The safety of these isolates further supports their application in functional foods and dietary supplements. Additionally, the multi-strain probiotic consortium demonstrated excellent performance in a simulated gut model, highlighting its potential for real-world applications.

Supplementary Information4 The online version contains supplementary material available at https://doi.org/10.1007/s12223-025-01244-w.

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Author contribution MNK, SB, AA, and NA conducted experimental study; MNK analyzed the data and drafted the manuscript; KL identified the isolates through Rep-Seq; and MI and SDN supervised the study, critically analyzed the manuscript, and improved the final version for potential publication.

Data availability Data will be available on request.

Declarations

Conflict of interest The authors have declared that no competing interest exists.

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